

```

1  #include <stdio.h>
2
3  void firstFit(int blocks[], int m, int process[], int n) {
4      int allocation[n];
5
6      for(int i = 0; i < n; i++)
7          allocation[i] = -1;
8
9      for(int i = 0; i < n; i++) {
10         for(int j = 0; j < m; j++) {
11             if(blocks[j] >= process[i]) {
12                 allocation[i] = j;
13                 blocks[j] -= process[i];
14                 break;
15             }
16         }
17     }
18
19     printf("\nFirst Fit Allocation:\n");
20     for(int i = 0; i < n; i++) {
21         if(allocation[i] != -1)
22             printf("Process %d -> Block %d\n", i+1, allocation[i]+1);
23     } else
24         printf("Process %d -> Not Allocated\n", i+1);
25     }
26 }
27
28 void bestFit(int blocks[], int m, int process[], int n) {
29     int allocation[n];
30
31     for(int i = 0; i < n; i++)
32         allocation[i] = -1;
33
34     for(int i = 0; i < n; i++) {
35         int bestIndex = -1;
36         for(int j = 0; j < m; j++) {
37             if(blocks[j] >= process[i]) {
38                 if(bestIndex == -1 || blocks[j] < blocks[bestIndex])
39                     bestIndex = j;
40             }
41         }
42
43         if(bestIndex != -1) {
44             allocation[i] = bestIndex;
45             blocks[bestIndex] -= process[i];
46         }
47     }
48
49     printf("\nBest Fit Allocation:\n");
50     for(int i = 0; i < n; i++) {

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50+     for(int i = 0; i < n; i++) {
51         if(allocation[i] != -1)
52             printf("Process %d -> Block %d\n", i+1, allocation[i]+1);
53         else
54             printf("Process %d -> Not Allocated\n", i+1);
55     }
56 }
57
58+ void worstFit(int blocks[], int m, int process[], int n) {
59     int allocation[n];
60
61     for(int i = 0; i < n; i++)
62         allocation[i] = -1;
63
64+     for(int i = 0; i < n; i++) {
65         int worstIndex = -1;
66+         for(int j = 0; j < m; j++) {
67+             if(blocks[j] >= process[i]) {
68                 if(worstIndex == -1 || blocks[j] > blocks[worstIndex])
69                     worstIndex = j;
70             }
71         }
72
73+         if(worstIndex != -1) {
74             allocation[i] = worstIndex;
75             blocks[worstIndex] -= process[i];
76         }
77     }
78
79     printf("\nWorst Fit Allocation:\n");
80+     for(int i = 0; i < n; i++) {
81         if(allocation[i] != -1)
82             printf("Process %d -> Block %d\n", i+1, allocation[i]+1);
83         else
84             printf("Process %d -> Not Allocated\n", i+1);
85     }
86 }
87
88+ int main() {
89     int m, n, choice;
90
91     printf("Enter number of memory blocks: ");
92     scanf("%d", &m);
93
94     int blocks[m];
95     printf("Enter sizes of memory blocks:\n");
96     for(int i = 0; i < m; i++)
97         scanf("%d", &blocks[i]);
98
99     printf("Enter number of processes: ");
100    scanf("%d", &n);

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101
102     int process[n];
103     printf("Enter sizes of processes:\n");
104     for(int i = 0; i < n; i++)
105         scanf("%d", &process[i]);
106
107     printf("\n1. First Fit\n2. Best Fit\n3. Worst Fit\n");
108     printf("Enter your choice: ");
109     scanf("%d", &choice);
110
111     // Copy original blocks for reuse
112     int temp[m];
113     for(int i = 0; i < m; i++)
114         temp[i] = blocks[i];
115
116     switch(choice) {
117     case 1:
118         firstFit(temp, m, process, n);
119         break;
120     case 2:
121         for(int i = 0; i < m; i++) temp[i] = blocks[i];
122         bestFit(temp, m, process, n);
123         break;
124     case 3:
125         for(int i = 0; i < m; i++) temp[i] = blocks[i];
126         worstFit(temp, m, process, n);
127         break;
128     default:
129         printf("Invalid choice!\n");
130     }
131
132     return 0;
133 }

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```
Number of memory partition : 5
Number of process : 4
Enter the memory partitions :
150
220
500
350
700
ENter process size :
160
450
500
412
1. Firstfit      2. Bestfit      3. worstfit
Enter your choice : 1

160 fits in 220
450 fits in 500
500 fits in 700
412 must wait for its process
-----
Process exited after 31.7 seconds with return
Press any key to continue . . .
```